Xu Yuanjian(徐元健)

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Education

Hong Kong University of Science and Technology, Ph.D. in Fintech 2023.09–2027.04
Peking University, Master in Computer Science 2020.09–2023.06
Nankai University, Bachelor in Computer Science 2014.09–2018.06

A Brief Introduction

I am a Ph.D. candidate at the Hong Kong University of Science and Technology, Guangzhou campus, supervised by Prof. Guang Zhang from HKUST-GZ and Dr. Zhong Li from MSRA. My past research has focused on **sample selection and sample quality assessment across the full LLM training lifecycle**. Recently, I have been exploring two directions:

- **Multimodal Data Selection:** Studying how data selection strategies should adapt to multimodal corpora with extremely low signal-to-noise ratios.
- **Specialized Text Data Selection:** Understanding what domain-specific data, such as agentic trajectories and long-context documents, contributes to LLM training, and developing principled criteria for selecting examples that provide useful capabilities and learning signals.

My work has been published at leading venues including **NeurIPS 2026**, **ICML 2026 (×2)**, **ACL 2026 (×2)**, **ACL 2025**, **ACL 2023**, **ICASSP 2026**, and **ACM ICAIF 2025**. I also serve as a reviewer for leading conferences such as **NeurIPS**, **ICLR**, and **ICML (🏆Gold Reviewer Award)**. I was selected for top talent internship programs at **Tencent Hunyuan**, **MSRA**, and **Tsinghua AIR**.

Published Papers

- [ICML 2026 (Spotlight 🏆)] **Yuanjian Xu**, et al. "Towards Efficient LLMs Annealing with Principled Sample Selection." **[CCF A] [CORE A*]** @ MSRA
- [ICML 2026] **Yuanjian Xu**, et al. "D^3: Dynamic Directional Graph-Constrained Data Scheduling for LLM Training." **[CCF A] [CORE A*]** @ MSRA
- [ACL 2026] **Yuanjian Xu**, et al. "Rethinking Data Mixing from the Perspective of Large Language Model." **[CCF A] [CORE A*]**
- [NeurIPS 2026] Yuxuan Sun, **Yuanjian Xu**, et al. "Rethinking Knowledge Distillation for Diffusion Language Models." **[CCF A] [CORE A*]** @ MSRA
- [ACL 2023] **Yuanjian Xu**, et al. "Hard Sample Aware Prompt-Tuning." **[CCF A] [CORE A*]**
- [ACL 2026] **Yuanjian Xu**, et al. "BizCompass: Benchmarking the Reasoning Capabilities of LLMs in Business Knowledge and Applications." **[CCF A] [CORE A*]**
- [ACL 2025] **Yuanjian Xu**, et al. "FinRipple: Aligning Large Language Models with Financial Market for Event Ripple Effect Awareness." **[CCF A] [CORE A*]**
- [ICASSP 2026 (Oral 🏆)] **Yuanjian Xu**, et al. "HGAN-SDEs: Learning Neural Stochastic Differential Equations with Hermite-Guided Adversarial Training." **[CCF B] [CORE A]**
- [ICAIF 2025] **Yuanjian Xu**, et al. "LENS: Large Pre-trained Transformer for Exploring Financial Time Series Regularities." (Leading conference for AI in finance)

Under Review

- [**Under Review @ ACL 2027**] **Yuanjian Xu**, et al. "Data Strategy in Action: Identifying High-Quality Corpora Across the LLM Training Lifecycle." **[CCF A]**
- [**Under Review @ ICLR 2027**] **Yuanjian Xu**, et al. "CUBE: Benchmarking Sample Reasoning Quality from Internal Geometric Dynamics in LLMs." **[CCF A]** @ Tencent Hunyuan
- [**Under Review @ ICLR 2027**] Jinyi Han, Zixiang Di, Zishang Jiang, Ying Liao, Jianing Hao, **Yuanjian Xu**, and Yanghua Xiao. "Structured Reasoning for Large Language Models." **[CCF A]**
- [**Under Review @ ACL 2027**] Jinyi Han*, **Yuanjian Xu***, et al. "Skill-Use: Can LLMs Actually Use Skills in Agentic Harnesses?" **[CCF A]** Equal contribution; @ Tencent Hunyuan
- [**Under Review @ ACL 2027**] **Yuanjian Xu**, et al. "Can LLMs Stay Robust in Noisy Agentic Environments?" **[CCF A]** @ Tencent Hunyuan
- [**Under Review @ ACL 2027**] Wei Guo, **Yuanjian Xu**, et al. "Bridging Foundational and Agentic Capabilities of LLMs." **[CCF A]**
- [**Under Review @ AAAI 2027**] **Yuanjian Xu**, et al. "Rethinking Neural SDEs under Shifting Data-Generating Processes." **[CCF A]**

Internship Experience

Tencent Hunyuan QingYun Program Research Intern | 2026.05–Present

Focused on reasoning and agentic data quality assessment for large language models.

Joinquant (Billion-scale quantitative fund) Research Intern | 2026.02–2026.05

Worked under the supervision of PM Ruixiao. Developed tick-level generative and representation models for high-frequency trading. Addressed key challenges including non-equally spaced data and market randomness.

Microsoft Research Asia (MSRA) Tomorrow Star Research Intern | 2025.10–2026.02

Worked under the supervision of Dr. Zhong Li. Focused on data selection and training order optimization for large language models. Proposed the D^3 and DiReCT methods, both accepted at ICML 2026.

AIR, Tsinghua University Research Intern | 2022.03–2023.03

Worked under the supervision of Prof. Zaiqing Nie. Contributed to the Meituan Nutrition Knowledge Graph construction. Investigated hard sample problems in NLP and proposed HardPT, published at ACL 2023.

Academic Service

Reviewer: ICML (2025, 2026), NeurIPS (2024, 2025, 2026), ICLR (2025, 2026), AAAI 2026, COLM 2026

Hong Kong University of Science and Technology, Teaching Assistant, Advanced Statistics

Skills

Machine Learning: I am well-versed in modern generative models, LLMs, and machine learning.
Mathematical Foundation: I have a relative solid mathematical foundation. During my Ph.D. studies, I passed the qualifying exam in optimization, advanced probability theory, advanced statistics, and stochastic analysis.
Financial Theory: I also have exposure to financial theory, having passed FRM Level 1 and studied advanced microeconomics and advanced asset pricing.

Honors & Awards

- ICML Gold Reviewer Award 2026
- QingYun Program (Top Talent Internship Program), Tencent Hunyuan 2026
- Tomorrow Star Plan (Top 10% Internship Program), Microsoft Research Asia 2025
- Full Ph.D. Scholarship, Hong Kong University of Science and Technology 2023–Present
- Award for Excellent Academic Excellence, Peking University 2021
- AIR Star Plan, Tsinghua University 2021